Week 4 - Windows Artefacts & Registry Forensics

One-Page Cheat-Sheet | Group 10 | Digital Forensics Internship

Core artefacts and what they prove

Artefact	Source hive / file	Registry path / location	What it tells	Common tool
RunMRU	NTUSER.DAT	Software\Microsoft\Windows\CurrentVersion\Explorer\RunMRU	Commands typed in Windows Run box	Registry Explorer / Python
RecentDocs	NTUSER.DAT	Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs	Recently opened files and file-type categories	Registry Explorer / Autopsy
UserAssist	NTUSER.DAT	...\Explorer\UserAssist\{GUID}\Count	GUI-based program execution traces; often ROT13 encoded	RegRipper / Python
TypedPaths	NTUSER.DAT	...\Explorer\TypedPaths	Folder paths typed in File Explorer	Registry Explorer
USBSTOR	SYSTEM	ControlSet001\Enum\USBSTOR	Connected USB storage device vendor, product and serial traces	Registry Explorer / USBDeview
MountedDevices	SYSTEM	MountedDevices	Drive-letter / volume mapping evidence	Registry Explorer
Installed Programs	SOFTWARE	Microsoft\Windows\CurrentVersion\Uninstall	Installed applications, version, publisher, install date	Registry Explorer / Python
Startup Programs	NTUSER/SOFTWARE	...\CurrentVersion\Run / RunOnce	Programs configured to start automatically	Autoruns / Registry Explorer
Prefetch	File system	C:\Windows\Prefetch	Application execution, run count and last run time	WinPrefetchView / PECmd
Recycle Bin	File system	C:\$Recycle.Bin	Deleted file name, original path and deletion information	Autopsy / ShellBags tools
Event Logs	EVTX files	C:\Windows\System32\winevt\Logs	Logon/logoff, system events, PowerShell and device events	Event Viewer / EvtxECmd

Quick investigation tips: Work on copies of evidence; record source path, timestamp and tool used; cross-check one finding with another artefact; export screenshots/CSV/PDF for reporting; do not rely on one artefact alone.

Short Summary of Week 4 Collective Findings

Week overview: The week focused on Windows artefacts and registry forensics. The objective was to create a shared understanding of key Windows evidence sources and prepare a reusable one-page reference for future investigations.

Available group output: The available parser output showed that NTUSER.DAT was loaded successfully and that user activity artefacts such as RecentDocs, UserAssist and UserStartup could be read. RecentDocs contained multiple values and subkeys, while UserStartup showed a OneDrive auto-start entry.

Important collective findings: Registry artefacts are useful for answering questions about user activity, USB/device history, installed software, startup entries and Windows configuration. However, registry evidence should be correlated with Prefetch, Event Logs, LNK files, Recycle Bin and timeline data for stronger conclusions.

Challenges faced: Main challenges include missing/empty keys, binary registry data that is not directly readable, locked hive files on live Windows systems, and the need to decode artefacts such as UserAssist and RecentDocs.

How to improve: Use exported hive files from FTK Imager or reg save, decode binary values, parse UserAssist Count keys, export structured JSON/CSV, and keep a consistent evidence note format for every finding.

Conclusion: The cheat-sheet can be used as a quick reference by all groups. It links each artefact to its source, path, purpose and tool so investigators can quickly decide where to look during Windows forensic analysis.

Final deliverables: Presentation + One-page Cheat-Sheet + Short Summary Report